$$\frac{a_x + a_M}{a_y} = \cot\alpha \tag{5}$$

Solving the above equations gives

$$N = \frac{(\cos\alpha + \mu\sin\alpha)mg}{1 + (m/M)\sin^2\alpha - \mu(m/M)\sin\alpha\cos\alpha} \tag{6}$$

On AB, replace α by β. Now decide whether the wedge moves, using (3). If it does, then (3) and (4) require

$$a_M = \frac{1}{M}(N\sin\alpha - \mu N\cos\alpha - \mu Mg) > 0 \tag{7}$$

Else the wedge stays still; the block slides down it.

On BC, $\alpha = 45°$, using (6)、(7) and the given data yields

$$N = 28.2\ \text{N} \tag{8}$$

$$a_M = 2.50 \times 10^{-2}\ \text{m}\cdot\text{s}^{-2} > 0 \tag{9}$$

On AB, use $\beta = 60°$ in (6)、(7) in place of $\alpha = 45°$; N' for N, a_M' for a_M. From (6)、(7) and numerical data,

$$N' = 20.1\ \text{N} \tag{8'}$$

$$a_M' = -3.90 \times 10^{-2}\ \text{m}\cdot\text{s}^{-2} < 0 \tag{9'}$$

Thus wedge rests from A to B, but moves from B to C. The block's work on the wedge is

$$W = N\sin\alpha \cdot s \tag{10}$$

s is the wedge displacement;

$$s = \frac{1}{2}a_M t^2 \tag{11}$$

Find t from vertical block motion from B to C, using (2) and (8):

$$a_y = g - \frac{N}{m}\cos\alpha = 5.01\ \text{m}\cdot\text{s}^{-2}$$

By kinematics,

$$h_2 = v_{0y}t + \frac{1}{2}a_y t^2 \tag{12}$$

v_{0y}: vertical part of v_0 just past B. Obtain v_0 by energy conservation:

$$\frac{1}{2}mv_0^2 = mg(h_1 - h_2) \tag{13}$$

Thus
$$v_0 = \sqrt{2g(h_1 - h_2)} = 4\ \text{m}\cdot\text{s}^{-1}$$

$$v_{0y} = v_0\cos\alpha \tag{14}$$

Use (14) in (12) to find

$$t = \frac{-v_{0y} \pm \sqrt{v_{0y}^2 + 2a_y h_2}}{a_y}$$

Take the positive sign and insert data:

$$t = 0.492\ \text{s}$$

Use (11), recalling (9):

$$s = \frac{1}{2}a_M t^2 = 3.01 \times 10^{-3}\ \text{m} \tag{15}$$

Use (10):

$$W = N \cdot \sin\alpha \cdot s = 6.00 \times 10^{-2}\ \text{J} \tag{16}$$